

Assignment - 3

1) Top-Down design

Specification



System design



Module design



RTL design



Simulation & Verification



Synthesis & Implementation



Testing

(2) Pipelining a 4-stage circuit

Nonpipelined

In $\rightarrow$ St 1 $\rightarrow$ St 2 $\rightarrow$ St 3 $\rightarrow$ St 4 $\rightarrow$ output

Pipelined

In $\rightarrow$ S₁ $\rightarrow$ reg $\rightarrow$ S₂ $\rightarrow$ reg $\rightarrow$ S₃ $\rightarrow$ reg $\rightarrow$ S₄ $\rightarrow$ reg $\rightarrow$ output

(3) resource sharing

without:

A+B $\rightarrow$ add 1

C+D $\rightarrow$ add 2

with:

A/B or C/D $\rightarrow$ mux $\rightarrow$ single adder $\rightarrow$ reg.

(4) pipelined & shared resource cu

Parameter	Effect
Speed	higher
Area	lower
Power	lower
latency	higher